



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer any **one** question on the answer paper provided.
Circle the question attempted on the cover page.

The use of an approved scientific calculator is expected, where appropriate.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/ 12
3	/ 12
4	/ 12
5	/ 12
6	/ 10
7	/ 12
Section B	
8 or 9*	/ 20
Total	

This Question Paper consists of **22** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. DNA polymerases carry out DNA replication during cell division, with tight coordination of leading and lagging strand synthesis. **Fig. 1.1** shows two representations of a bacterial DNA polymerase, DNA Pol III: a space-filling model and a ribbon diagram.

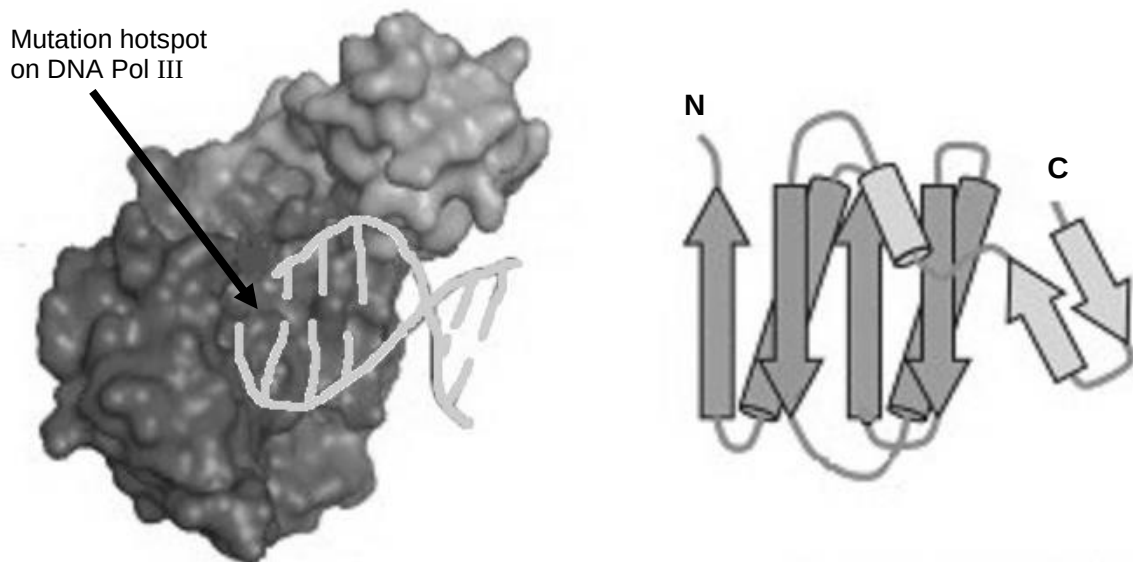


Image adapted from Cell, Vol 126, Issue 5

Fig. 1.1

- (a) Describe the structure of DNA Pol III. [2]
1. Comprise a single polypeptide chain
 2. folded into (antiparallel) β -pleated sheets / α -helices
 3. Further folded into spherical / globular
 4. three-dimensional conformation
- (b) Explain the significance to its activity of the structure of DNA Pol III. [2]
1. Gives rise to an active site
 2. whose shape is complementary
 3. to the DNA nucleotide sequence
 4. on the DNA origin of replication
- (c) DNA Pol III is often used in laboratory for the synthesis of DNA under high temperature. Suggest one structural property that makes it a suitable [1]

molecule for such processes.

Extensive covalent bonds / disulphide bridges between R groups of amino acid residues

- (d)** The arrow in **Fig. 1.1** shows a mutation hotspot on DNA Pol III where substitution mutation is common.

State two mutations that would not affect the activity of DNA Pol III. Explain your answer.

[5]

1. **Silent mutation**
2. **Change the corresponding codon on mRNA**
3. **Does not change the corresponding amino acid on polypeptide chain**
4. **due to degeneracy of the genetic code/ different codon code for the same amino acid**
5. **Folding of the polypeptide chain unaffected**
6. **Three-dimensional conformation of the active site unchanged**

7. **Missense mutation**
8. **Changes corresponding codon on mRNA**
9. **Changes corresponding amino acid on polypeptide chain;**
10. **to one with similar property**
11. **Binding to DNA unaffected**

[Total: 10]

2. The galactose operon (*gal* operon) controls the utilization of galactose in *Escherichia coli*. The operon consists of three genes (*galE*, *galT* and *galK*) that code for enzymes involved in breaking down of galactose (Gal) into UDP-galactose (UDP-Gal) and glucose-1-phosphate (Glu-1-P), which can be used for cell wall synthesis and cellular metabolism.

Similar to the gene regulation of *lac* operon, the *gal* operon is also under dual control: negative control by the *gal* repressor, which is encoded by *galR* gene and positive control by cAMP receptor protein (CRP), which is activated by cAMP in the absence of glucose.

Fig. 2.1 shows the structure of a *gal* operon and the biochemical pathway involving the enzymes encoded by the *gal* operon.

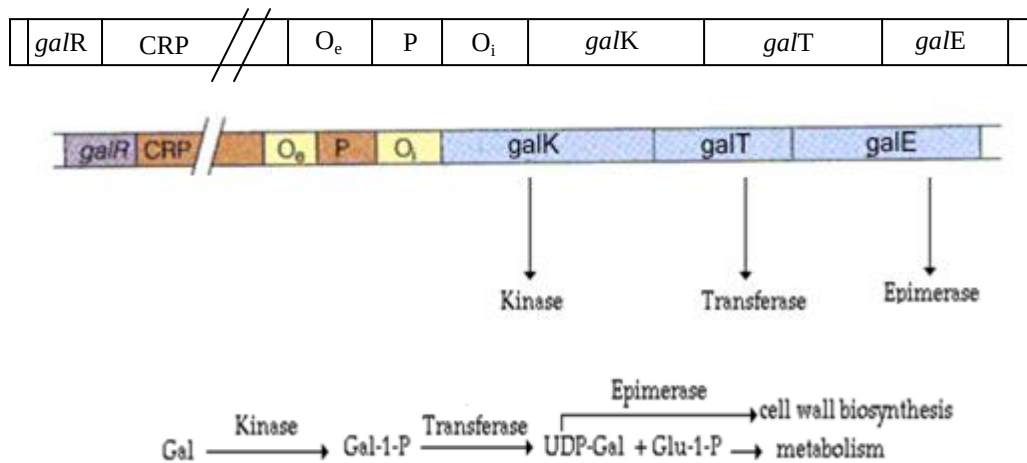


Image adapted from <http://www.pearsonhighered.com> & <https://www.bio.cmu.edu>

Fig. 2.1

(a) With reference to **Fig. 2.1**,

- (i) describe how the *gal* operon is regulated when galactose is added to an *E.coli* suspension. [2]

1. galactose binds to gal repressor
2. gal repressor is inactivated
3. leaves/ does not bind to the operator
4. allows RNA polymerase to bind to promoter to initiate transcription

- (ii) describe the change in regulation of the *gal* operon when glucose is added to the suspension in (a)(i). [2]

1. decreases concentration of cAMP
2. more CRP remains inactive/ fewer CRP-cAMP complexes
3. RNA polymerase has a lower affinity to promoter
4. Rate of transcription is low

- (b) In a mutated strain of *E.coli*, the bacteria can use galactose for metabolism but was unable to synthesise cell wall due to a mutation in the *gal* operon.

State and explain the location of the mutation.

[3]

1. Mutation is within *galE* gene
2. epimerase is non-functional/ not synthesized
3. UDP-galactose is not converted to cell wall component
4. *galK* and *galT* genes are normal/ kinase and transferase are functional
5. indicates that regulatory sequences are normal

λ bacteriophage is known to infect *E.coli* and can undergo both lytic and lysogenic cycles in the host cells. The bacteriophage inserts its own genome between the *gal* operon and *bio* operon.

Fig. 2.2 shows how the λ bacteriophage infects *E.coli*. However, an error occurred during the excision of phage DNA.

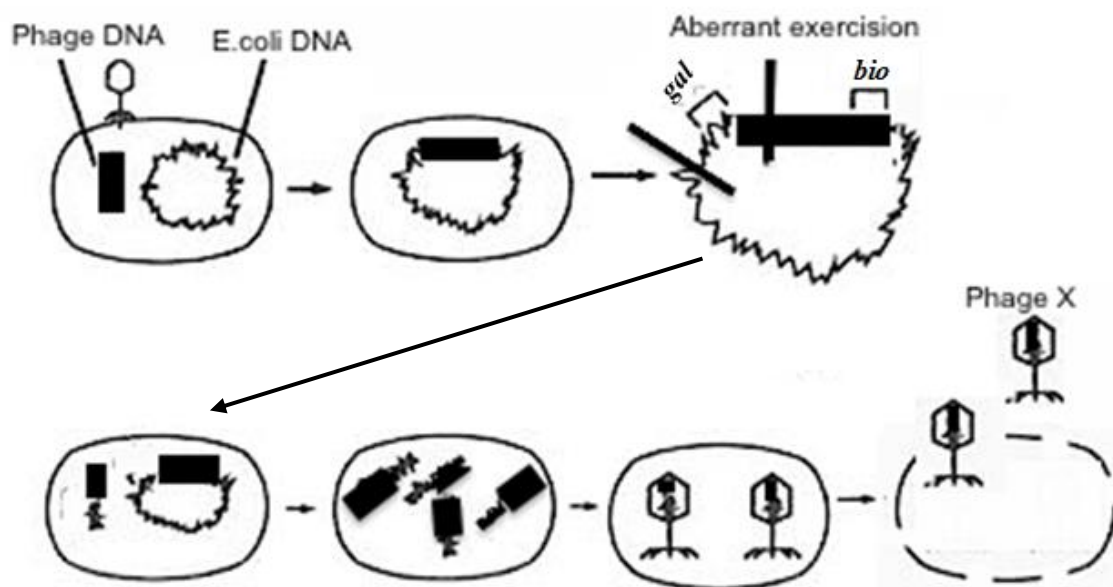


Image adapted from <http://medchrome.com>

Fig. 2.2

- (c) Compare how lytic cycle differs from lysogenic cycle.

[3]

Factors	Lysogenic	Lytic
Formation of prophage	Yes	No
Viral genome	Viral genome is <u>incorporated</u> into host genome	Viral genome <u>lies in the cytoplasm</u>

Expression of genes	<u>One</u> prophage gene coding for λ repressor protein is expressed during lysogenic cycle	<u>Many</u> viral genes coding for phage enzymes and structural components are expressed during lytic cycle
Replication of viral genome	<u>Replicated together</u> with bacterial DNA and passed to daughter cells	<u>Replicated independently</u> and packaged into new viral progeny
Bacterial DNA	<u>Replicated together</u> with phage DNA and passed to daughter cells during lysogenic cycle	<u>Hydrolysed</u> during lytic cycle
Fate of host cell	<u>Survive</u> to produce daughter bacterial cells during the lysogenic cycle	<u>Killed</u> upon viral release during lytic cycle

(d) With reference to **Fig. 2.2**, suggest why the mutated strain of *E.coli* can regain its cell wall synthesis function if it is infected by the Phage X.

[2]

1. Improper excision of viral DNA to include bacterial *gal* operon
2. Mutated *gal* operon replaced by functional *gal* operon from previous host
3. by recombination/ incorporation into bacterial DNA/ homologous DNA exchange

[Total: 12]

3.

Part I

Fig. 3.1 is an electron micrograph of a cell.

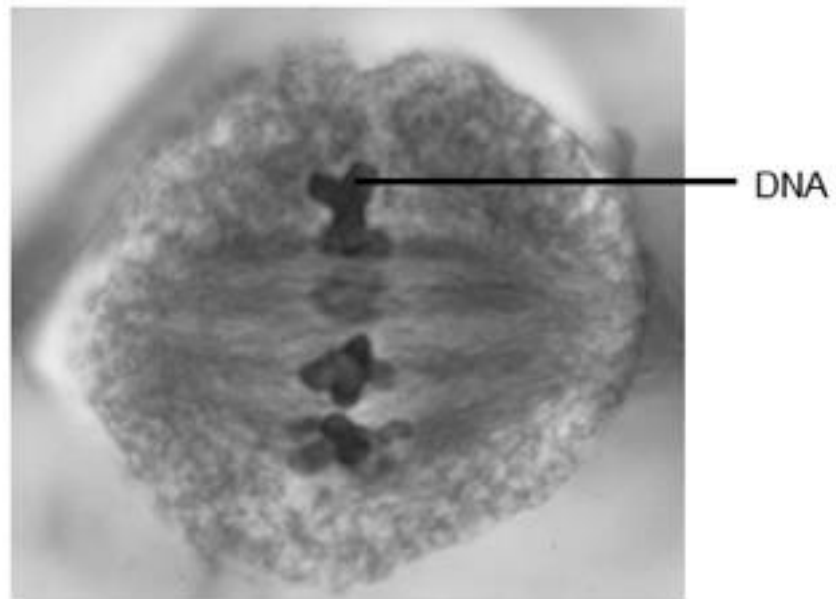


Image adapted from Carolina Biological Supply Company

Fig. 3.1

- (a) Describe how the DNA shown in **Fig. 3.1** is folded and the resultant impact on its transcriptional activity.

[3]

1. DNA molecule form nucleosomes / beads-on-string structure
2. further coiling to form 30nm fibre / solenoid
3. Interacts with scaffold proteins
4. and further condensed into chromosome
5. Promoter is not accessible to RNA polymerase
6. Transcription cannot be initiated

- (b) Describe how else transcriptional activity of a cell can be prevented.

[2]

Any 2

1. Addition of methyl groups to DNA
2. Removal of acetyl groups from histone tails
3. Addition of methyl groups to histones tails

Part II

Cells rely on chaperone proteins to prevent nearby proteins from interfering with protein folding. Protein folding can occur in the cytoplasm and proteins must fold into defined three-dimensional structures to gain functional activity. As protein molecules are highly dynamic, constant chaperone surveillance is required to ensure proper functioning of proteins.

(c) Explain how chaperone proteins help in the regulation of gene expression. [3]

- 1. Post-translational modification**
- 2. Through folding of polypeptide chain**
- 3. Increase the amount of functional protein (in the cytoplasm)**

(d) Suggest the advantage of regulating gene expression at this level. [1]

- 1. Allows for fast response to changes in environmental conditions**

(e) Describe how gene expression can be down-regulated at the protein level. [3]

- 1. Tagging of protein via ubiquitin**
- 2. recruits proteosome**
- 3. to degrade protein**
- 4. Decreasing amount of functional proteins in the cytoplasm**

[Total: 12]

4.

PART I

(a) Explain what is meant by the following terms:

(i) heterozygous

[1]

1. having two different alleles

2. for a gene

(ii) genotype

[1]

set of alleles present in an organism

Rickets is a childhood disorder involving the softening and weakening of bones. It is usually caused by a lack of vitamin D, calcium ions or phosphate ions. A rare form of rickets that cannot be successfully treated with vitamin D therapy is caused by a mutant allele on the X chromosome.

Fig. 4.1 shows a pedigree for a family that has a history of this condition.

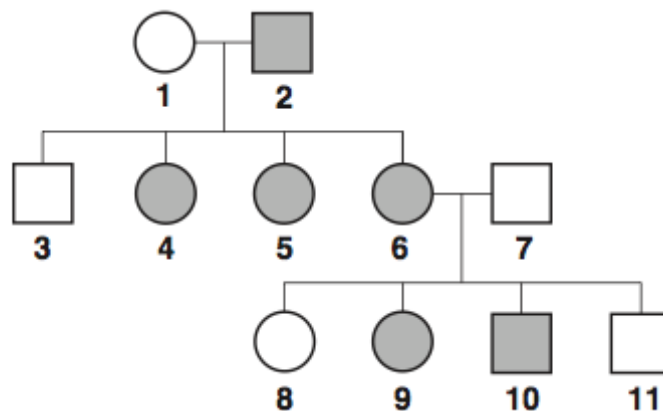


Fig. 4.1

(b) Using the symbols,

X^R for the mutant allele on the X chromosome;

X^r for the non-mutant allele on the X chromosome,

state the genotypes of the following individuals.

[3]

(i) Individual 1: X^rX^r

(ii) Individual 3: X^rY

(iii) Individual 9: X^RX^r

4.

PART II

The budgerigar, *Melopsittacus undulatus*, shown in **Fig. 4.2**, is a parrot that is native to Australia.



Image adapted from University of Cambridge International Examinations, 2012

Fig. 4.2

A budgerigar can have blue, green, yellow or white feathers. Two genes, *A/a* and *D/d*, are involved in the inheritance of feather colour in budgerigars.

- A bird which has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant allele **A** and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

(c) A student crossed two green-feathered budgerigars.

Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers is 25%.

[5]

Parental phenotype: **Green feathers** **Green feathers**
Parental genotype: **AaDd** **AaDD**

Gametes:



Punnett Square to show fusion of gametes produced by the F_1 generation:

	AD	Ad	aD	ad
AD	AADD Green feathers	AADd Green feathers	AaDD Green feathers	AaDd Green feathers

aD	AaDD Green feathers	AaDd Green feathers	aaDD Yellow feathers	aaDd yellow feathers
----	------------------------	------------------------	-------------------------	-------------------------

Phenotypic ratio: 1 (yellow feathers) : 3 (green feathers)

(d) Explain how the two genes interact to give rise to the different feather colours. [2]

1. Allele A codes for an enzyme which converts precursor to blue pigments in the feathers
2. Allele D codes for an enzyme which converts precursor to yellow pigments in the feathers
3. Green feathers is due to presence of blue and yellow pigments/ combined effect of allele A and D
4. Absence of alleles A and D, precursor not converted to colour pigments so feathers appear white

[Total: 12]

5. Cathy investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. She used the apparatus shown in **Fig. 5.1**.

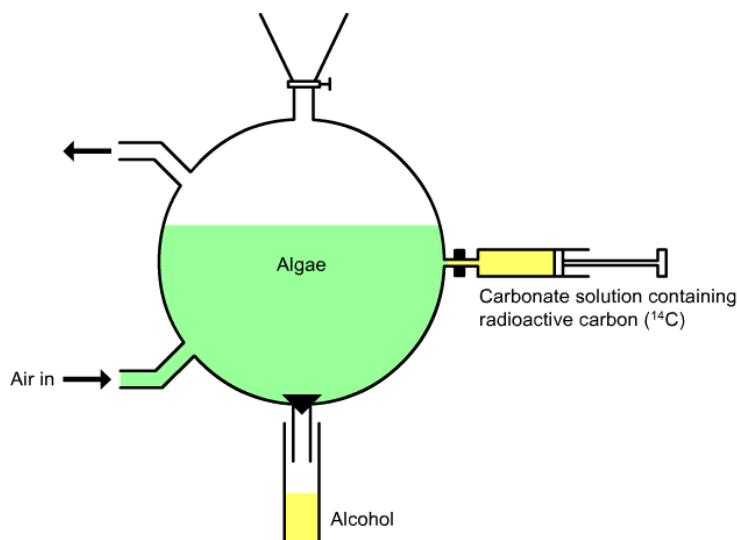


Image adapted from <http://www.snabonline.com>

Fig. 5.1

While the apparatus was in the dark room, Cathy supplied the algal cells with $^{14}\text{CO}_2$. The contents of the apparatus were thoroughly mixed and light was switched on subsequently. At five-second intervals, she released a few of the cells into hot alcohol, which killed the cells very quickly. The intermediates of the reactions were subsequently analysed and the results were shown in **Fig. 5.2**.

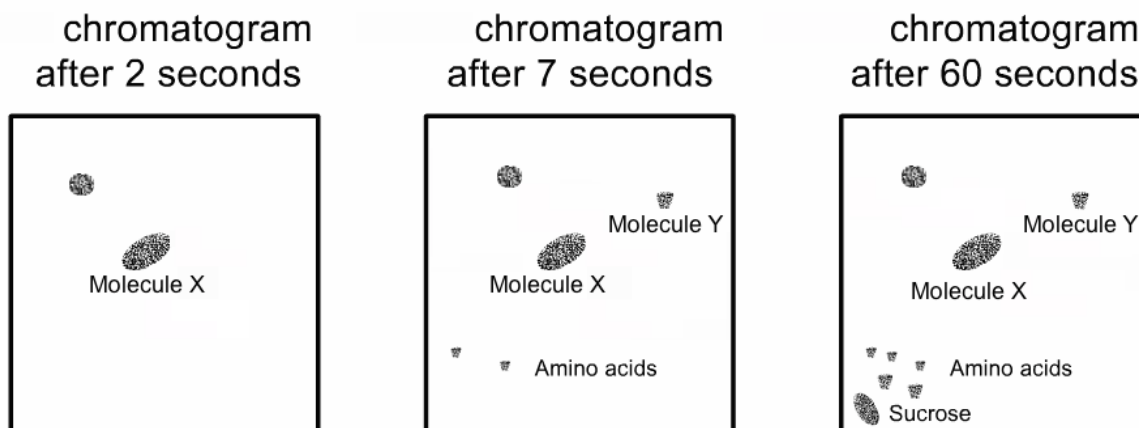


Image adapted from <http://www.snabonline.com>

Fig. 5.2

- (a) Describe how Molecule X and Molecule Y are formed.

[4]

Molecule X:

1. carbon fixation/ carboxylation
2. CO_2 is fixed with ribulose-1,5-bisphosphate
3. catalysed by ribulose-1,5, bisphosphate carboxylase
4. 6C intermediate is unstable
5. broken down into Molecule X-glycerate-3-phosphate

Molecule Y:

6. glycerate-3-phosphate is phosphorylated
7. into 1,3-bisphosphoglycerate
8. reduction
9. by NADPH
10. into triose phosphate

Dinitrophenol is a metabolic poison that can embed within the thylakoid membranes of chloroplasts and provide an alternate route for H^+ to diffuse across the thylakoid membranes.

- (b) Explain how the concentration of intermediates of Calvin cycle is affected by dinitrophenol. [2]

1. Less H^+ diffuse through ATP synthase
2. Less ATP produced for Calvin cycle
3. Accumulation of glycerate-3-phosphate
4. RuBP decreases
5. 1,3-bisphosphoglycerate/ glycerate-1,3-bisphosphate decreases

A further experiment was carried out by another student to determine the concentration of carbon dioxide in the leaves of plants at different times of the day. The results are shown in **Table 6.1**.

Table 6.1

Mean carbon dioxide concentration (ppm)	
8pm to 4am	8 am to 4pm
328	106

- (c) Explain how Calvin cycle and Krebs cycle can account for the difference in concentration of carbon dioxide in the leaves for the two periods shown in **Table 6.1**. [4]

8am to 4pm

1. Calvin cycle and Krebs cycle occur

8pm to 4am

2. No light
3. Calvin cycle stops, Krebs cycle occurs
4. More carbon dioxide released by cells compared to 8am to 4pm, vice versa
5. CO_2 used in Calvin cycle
6. During carbon fixation;

7. CO₂ released from Krebs cycle
8. Via oxidative decarboxylation

(d) Explain why higher concentration of carbon dioxide would increase carbon fixation during the period 8am to 4pm.

[2]

1. Increase amount of substrate -CO₂
2. Increase frequency of effective collision between CO₂ and active site of Rubisco
3. More enzyme-substrate complexes formed per unit time
4. More products (i.e. glycerate-3-phosphate) former per unit time

[Total: 12]

6.

Part I

In recent years, significant advances have been made in defining the regulatory pathways that control normal and abnormal cardiac valve development. One of such includes the receptor tyrosine kinase activation and downstream signaling pathway shown in **Fig. 6.1**. The epidermal growth factor (EGF) activates the MAPK/ERK pathway to induce the transcription of targeted gene.

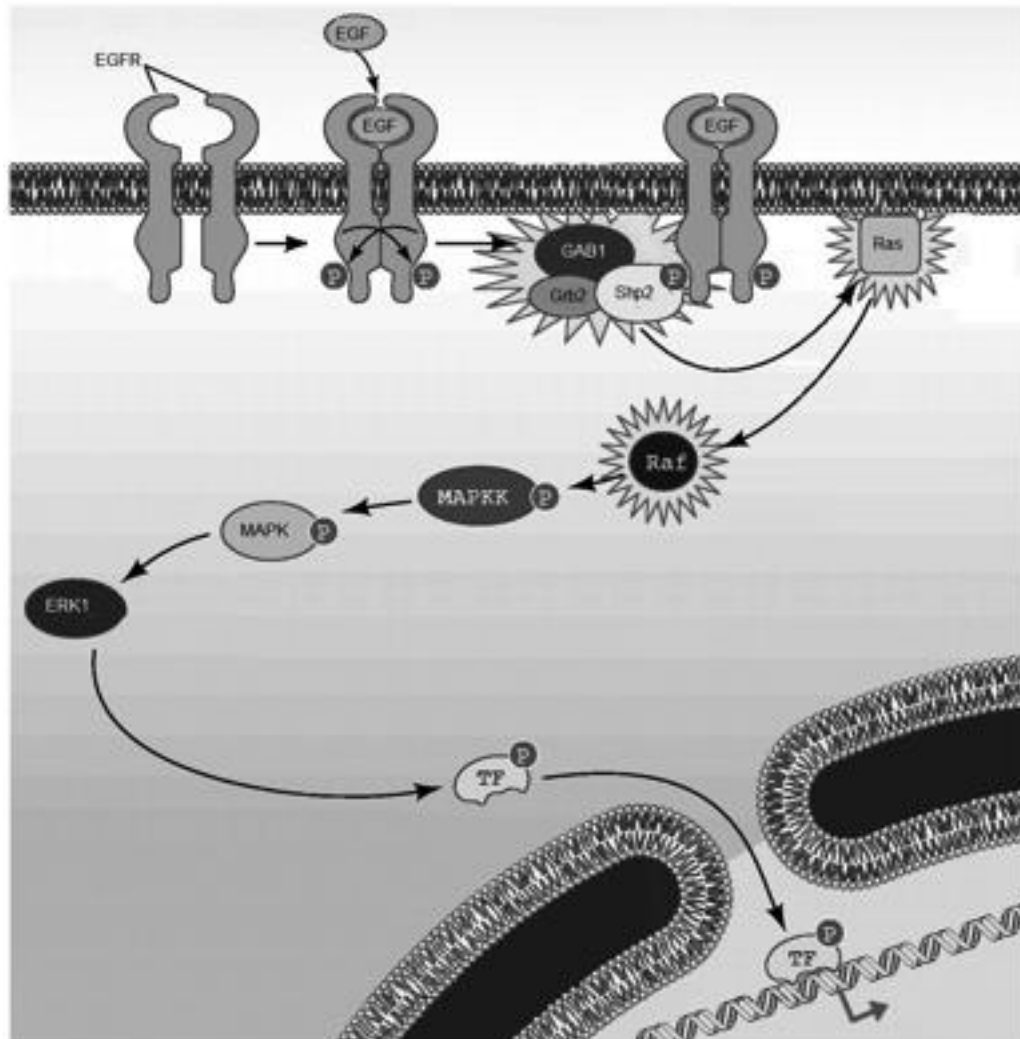


Image adapted from <http://physiologyonline.physiology.org/content/20/6/390/F1>

Fig. 6.1

(a) With reference to **Fig. 6.1**,

(i) describe how EGFR is activated,

[3]

1. EGF binds
2. EGFR binding site
3. via complementary shape
4. triggers dimerisation of EGFR subunits
5. This cause the subunits to undergo cross phosphorylation

6. at their tyrosine residues

- (ii) describe how the signal from the activated receptor is transduced to Raf. [3]

1. Activated receptor binds to GAB1-Grb2-Shp2 complex
2. phosphorylating / activating the complex
3. Complex dissociates from the EGFR
4. binds and activates Ras
5. which dissociate from the cell membrane
6. binds to and activate Raf

6.

Part II

Oral glucose tolerance test is often employed to diagnose diabetes when simple blood and urine tests give inconclusive results. Following a 12 hour fast, a blood sample is then taken to measure the patient's fasting blood glucose level. He is then given a drink containing 75g of glucose, and his blood glucose concentration tracked for the next 2 hours.

Fig. 6.2 shows the blood glucose concentration of two individuals, one healthy and one with diabetes.

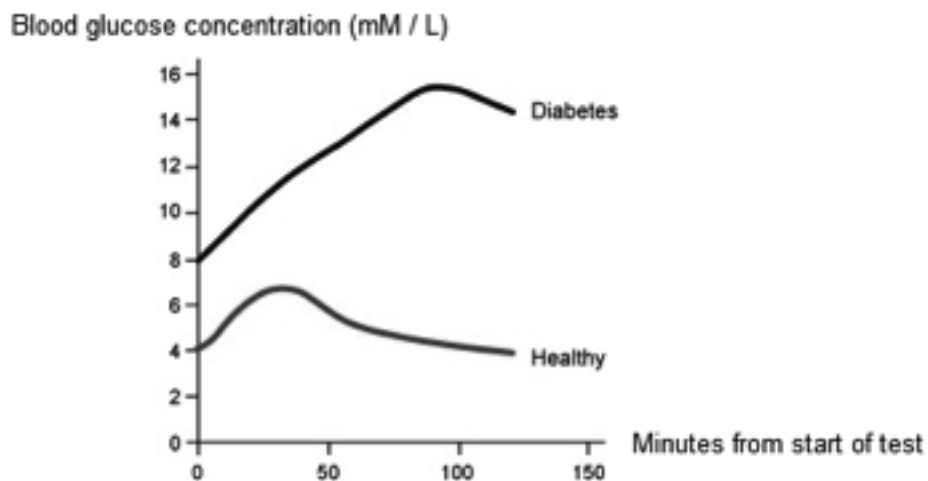


Image adapted from ABPI schools

Fig. 6.2

- (b) Describe and explain the results of the test for the diabetic patient in the first 100 minutes. [3]

1. From 0 min to 100 min, blood glucose concentration increases from 8 mM/L to 15.5 mM/L
2. Glucose is absorbed into the blood stream
3. and is not taken up by liver / muscle cells
4. Insulin is not secreted
5. by β cells of Islet of Langerhans of the pancreas

- (c) Suggest a reason for the drop in blood glucose concentration for the diabetic patient.

[1]

Glucose used for cellular respiration

[Total: 10]

7. Many species of anole lizards are found throughout the Caribbean and the surrounding mainland. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida.

Fig. 7.1 shows the distribution of four species of anole lizards.

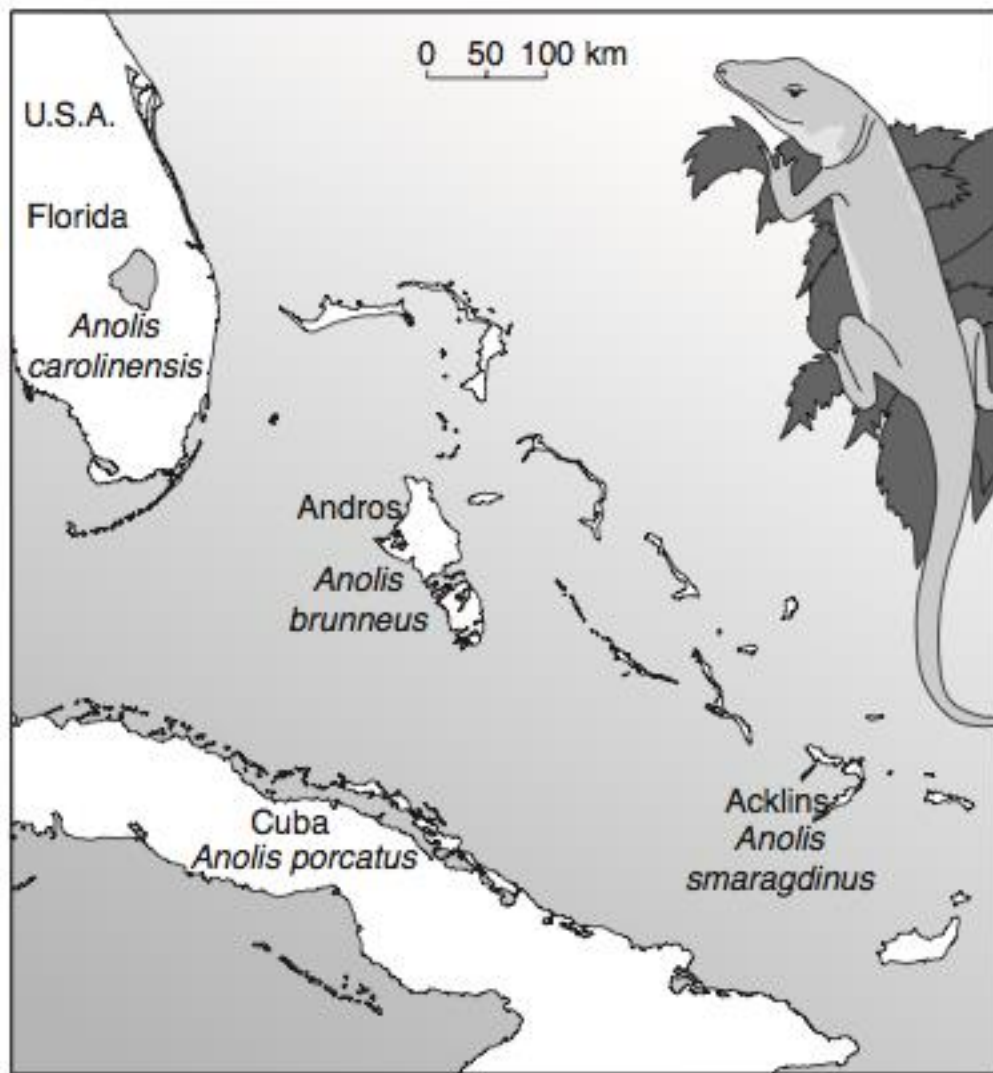


Image adapted from University of Cambridge International Examinations, 2010

A student carried out DNA analysis to investigate the relationships between these four lizards species. The base sequences of a region of mitochondrial DNA (mtDNA) from the four species were compared. **Table 7.1** shows the difference in base sequence of the species.

Table 7.1

	<i>A. brunneus</i>			
<i>A. brunneus</i>		<i>A. smaragdinus</i>		
<i>A. smaragdinus</i>	12.1		<i>A. carolinensis</i>	
<i>A. carolinensis</i>	16.7	15.0		<i>A. porcatus</i>
<i>A. porcatus</i>	11.3	8.9	13.2	

The student put forward the hypothesis that the three species, *A. brunneus*, *A. smaragdinus* and *A. carolinensis*, have originated from three separate events in which a few individuals of *A. porcatus* spread directly from Cuba to three different places.

- (a) Suggest why changes in mtDNA can be used to support the neutral theory of molecular evolution in closely related species. [3]

1. Constant rate of mutation of mtDNA
2. Small changes in mtDNA
3. mutations of base sequence on mtDNA are silent
4. no effect on the protein/ phenotype
5. phenotype is selectively neutral
6. changes in mtDNA may be lost/ fixed in the population by genetic drift
7. rather than natural selection

- (b) Explain how the results in **Table 7.1** support their hypothesis. [3]

1. *A. brunneus* and *A. porcatus* = 11.3
2. *A. smaragdinus* with *A. porcatus* = 8.9
3. *A. carolinensis* with *A. porcatus* = 13.2
4. Ref. to all 3 species and *A. porcatus* - Lowest difference in base sequence
5. more closely related to *A. porcatus* than to each other
6. supports hypothesis that *A. porcatus* is the common ancestor
7. suggests founder's effect

- (c) Suggest why the mtDNA of *A. porcatus* has the least differences with *A. smaragdinus* compared to *A. brunneus* and *A. carolinensis*. [1]

1. Speciation occurs more recently
2. Similar selection pressure

- (d)** Explain how a population of *A. porcatus* that became isolated on an island could evolve into a new species.

[5]

1. (Genetic) variations within the population
2. due to mutations
3. Environmental factors on the island act as selective pressures
4. Some individuals are better adapted to the environment/ selective advantage
5. survive, reproduce and pass on alleles to offspring
6. Some individuals are unable to adapt to the environment/ selective disadvantage
7. Will not survive and reproduce and pass on alleles to offspring
8. Change in allele frequencies in the population
9. Geographical isolation/ separated by water
10. No gene flow/ interbreeding between populations
11. Allopatric speciation
12. Genetic drift/ founder's effect

[Total: 12]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. **(a)** Compare cellulose and collagen.

[8]

Similarities

1. Both are polymers/ made up of many monomers
2. Both contain cross-linking between chains
3. Both confer high tensile strength
4. Both have fibrous structure
5. Both are insoluble in water
6. Both have hydrogen bonds

Differences

Features	Cellulose	Collagen
7. Monomer	β -glucose	Amino acid
8. Arrangement of monomer	Alternate monomer is inverted	Alternate monomer is upright;
9. Bond between the monomers	β 1,4 glycosidic bond	Peptide bond
10. Type of cross links	<u>Hydrogen bonds</u> between -OH groups of neighbouring chains	<u>Hydrogen bonds</u> between -NH group and -CO group of the polypeptide chains and <u>covalent bonds</u> between the carboxyl end and amino end of tropocollagen
11. Arrangement of chains	Not staggered	Staggered

(b) Explain why animals store triglycerides in addition to glycogen.

[5]

1. store more energy per unit mass than glycogen
2. contain large amounts of hydrogen atoms in their hydrocarbon chains (which can be oxidised during cellular respiration)
3. source of metabolic water

4. contain large amounts of hydrogen atoms in their hydrocarbon chains, (which can be oxidized to water)
5. confer buoyancy for aquatic animals
6. less dense than water
7. provide thermal insulation
8. poor conductor of heat
9. provide electrical insulation
10. as component of myelin sheath (which allows rapid transmission of impulse)
11. forms a natural buffer to absorb shock
12. helps to cushion fragile internal organs (from impact and physical damage)

(c) Relate the structure of mitochondria to its role in oxidation of glucose.

[7]

Structure	Function
1. Double membrane 2. Outer membrane act as a barrier	3. to separate the processes of cellular respiration from other cytosolic processes
4. Infolding of inner membrane to form cristae	5. Large surface area of inner member for the embedment of; 6. ATP synthase 7. for phosphorylation of ADP to ATP 8. Electron carriers 9. transfer electrons down electron transport chain 10. Proton pumps 11. for active transport of proton across the membrane
12. Inner membrane is impermeable to protons	13. Protons accumulate in the intermembrane space 14. Build up of proton gradient for chemiosmosis
15. matrix contains enzymes and substrates	16. for link reaction 17. and Krebs cycle

[Total: 20]

9. (a) Describe how the structure of a cholinergic synapse facilitates impulse transmission.

[9]

	Structure	Function
1	Synaptic vesicles in synaptic knob	Contains acetylcholine for impulse transmission
2	<u>Voltage-gated</u> Ca^{2+} channels in pre-synaptic membrane	Influx of Ca^{2+} triggers exocytosis
3	<u>Voltage-gated</u> Na^+ channels in pre-synaptic membrane	Influx of Na^+ depolarizes membrane
4	Choline transporter in pre-synaptic membrane	Takes in choline for synthesis of acetylcholine
5	Acetylcholinesterase in synaptic cleft	Breakdown acetylcholine to stop impulse transmission
6	Synaptic cleft is narrow	Allows for acetylcholine to move quickly across by diffusion
7	Na^+/K^+ pump in post-synaptic membrane	Restores resting membrane potential
8	<u>Chemically-gated</u> Na^+ channels in post-synaptic membrane	Diffusion of Na^+ depolarises membrane / generates EPSP
9	Synaptic vesicles only in presynaptic knob	Ensures unidirectional transmission of impulse
10	Chemically-gated Na^+ channels only in post-synaptic membrane	Ensures unidirectional transmission of impulse
11	Synaptic membrane is impermeable to ions	Allows for build up of membrane potential
12	Presence of mitochondria	Synthesise ATP for exocytosis/synthesis of acetylcholine/active transport
13	AVP	

- (b) Explain the cause of refractory period and its significance to impulse transmission.

[5]

1. Refractory period occurs due to hyperpolarization; of the axon membrane
2. Slow-moving gates of voltage-gated K^+ channels
3. results in excessive efflux of K^+ out of the axon
4. causing the membrane potential to fall below resting potential
5. No stimulus can be generated during the refractory period
6. Ensures impulse can only be transmitted in one direction along the axon
7. Prevents fatigue at synapses caused by overstimulation
8. Prevents damage to effectors caused by overstimulation
9. AVP

(c) Outline the function of cell membranes.

[6]

1. Boundary between intracellular and extracellular environment
2. maintains a distinct internal environment for cell function
3. Regulates movement of substances into and out of cell
4. selective barrier to large, polar and charged substances
5. Allows cells to interact / adhere together
6. via glycoproteins and glycolipids in plasma membrane
7. For signal transduction
8. contains proteins for signal reception
9. Site for formation of multi-enzyme complexes
10. to increase rates of enzyme-catalysed reactions
11. Compartmentalisation within cell
12. provides localized conditions for specific process
13. Maintain high concentrations of enzymes or substrates
14. allows incompatible reactions to be carried out simultaneously
15. AVP

[Total: 20]